

Chapter 4 Channel Capacity and Channel Coding Theorem

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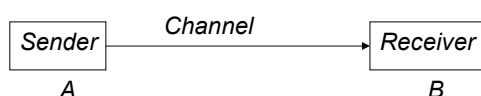
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Introduction



□ Communications Model (通信模型)

- What do we mean when we say that *A* communicates with *B*?



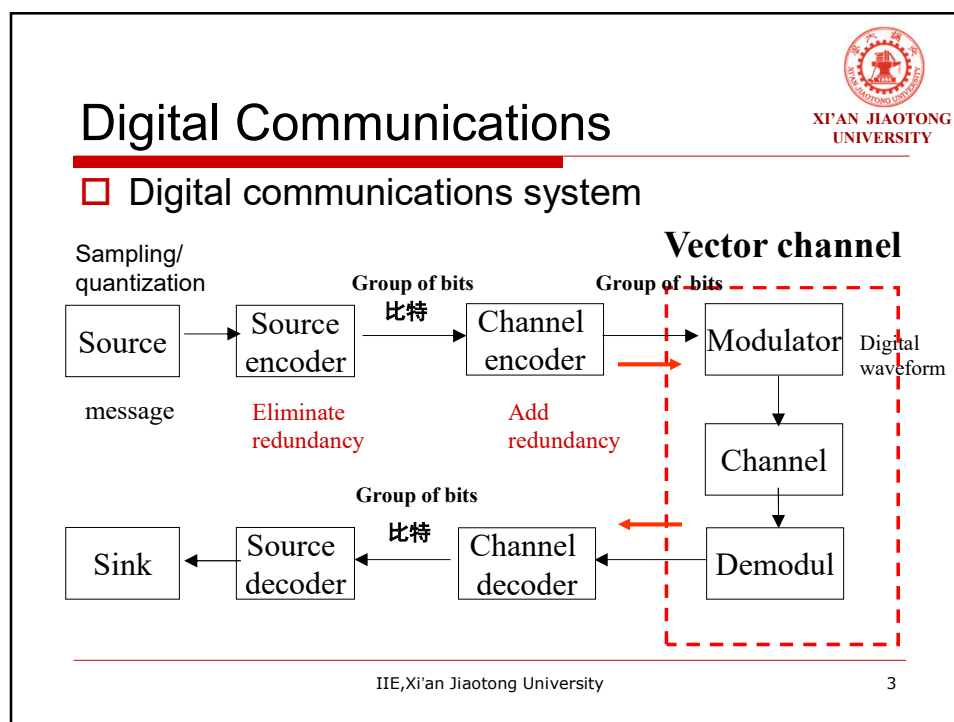
- The physical acts of *A* have induced a desired physical state in *B*. (A的物理变化带来了B的物理状态的变化)
- This transfer of information is a physical process and is subject to the uncontrollable noise and imperfections of the physical signaling process itself. (不可控噪声和信号非理想性)
- These imperfections makes that different input may give rise to the same output sequence, and therefore the inputs are confusable. (造成对发射信号的判断错误)
- We want to know the maximum number of distinguishable signals for *one use of* a communication channel.

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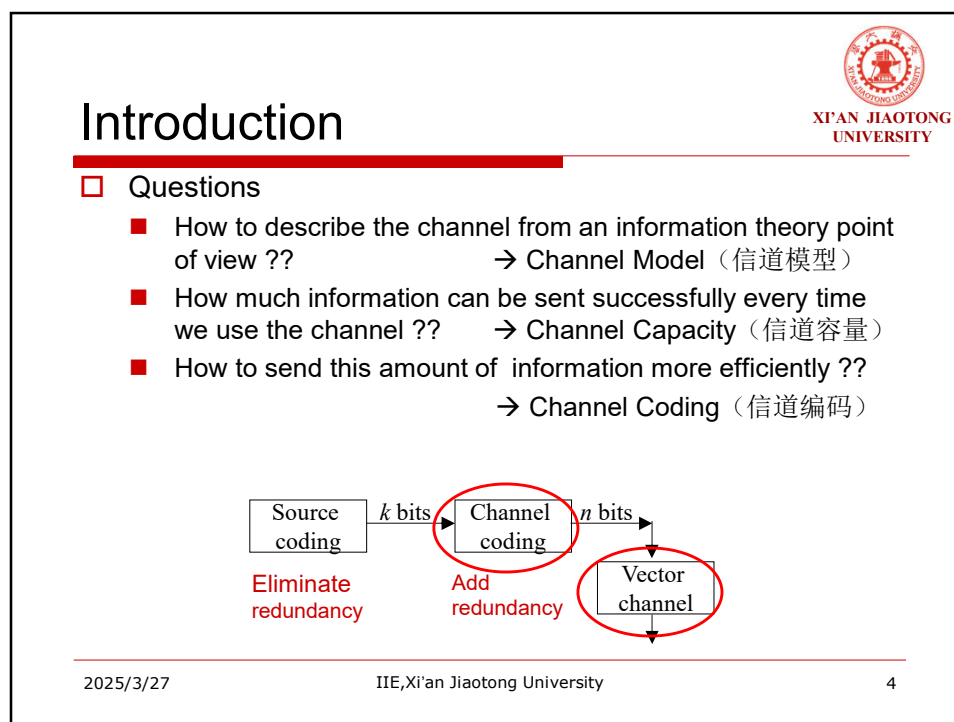
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Outline

- Channel Model
- Channel Capacity
- Channel Coder and Decoder
- Joint Typical Set
- Channel Coding Theorem (Shannon)

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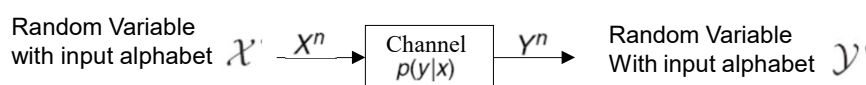
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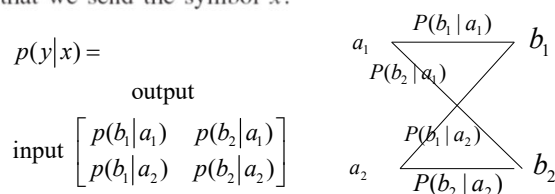
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Channel Model

- Probability model from an information theory point of view



Definition We define a *discrete channel* to be a system consisting of an input alphabet $\mathcal{X}$ and output alphabet $\mathcal{Y}$ and a probability transition matrix $p(y|x)$ that expresses the probability of observing the output symbol y given that we send the symbol x .



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Channel Model

Definition The n th extension of the discrete channel is the channel $(\mathcal{X}^n, p(y^n|x^n), \mathcal{Y}^n)$.

- Memoryless channel (无记忆信道) :
 - If the probability distribution of the output depends only on the input at that time and is conditionally independent of previous channel inputs or outputs.
- Memory channel (记忆信道)
 - If the probability distribution of the output depends not only on the input at that time but also on the previous time.
- Memoryless makes the channel easy to analyze, since the output only depends on the current input. Sequence transmission can be decoupled into symbol transmissions.

$$p(y^n|x^n) = \prod_{i=1}^n p(y_i|x_i)$$

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Channel Model

□ Some examples

- Noiseless binary channel (无噪信道)

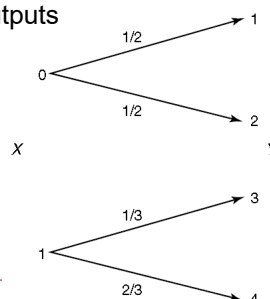
$$p(y|x) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

0 $\longrightarrow$ 0

1 $\longrightarrow$ 1

- Noisy channel with non-overlapping outputs

$$p(y|x) = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$



It appears to be noisy, but really is not.

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Channel Model

- 无噪信道与无损信道
- 有噪无损信道：信道转移矩阵每一列有一个且仅有一个非零元素

$$p(y|x) = \begin{bmatrix} a & & \\ & b & c \\ & & d \end{bmatrix}$$

噪声熵 $H(Y|X) \neq 0$

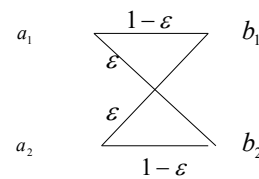
损失熵 $H(X|Y) = 0$

- Binary Symmetric Channel (BSC)

□ $\varepsilon = 0 \rightarrow$ Noiseless binary channel

□ $\varepsilon = 1/2 \rightarrow$ Strong noisy channel

$$p(y|x) = \begin{bmatrix} 1-\varepsilon & \varepsilon \\ \varepsilon & 1-\varepsilon \end{bmatrix}$$



$$H(X|Y) = H(X) - I(X;Y)$$

信息量的损失

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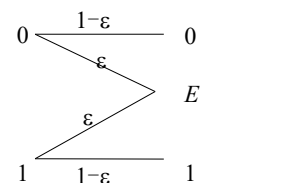
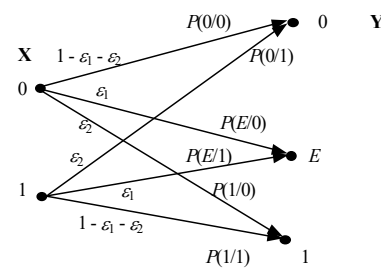
Channel Model

- Binary Erasure Channel

$$p(y|x) = \begin{bmatrix} 1-\varepsilon_1-\varepsilon_2 & \varepsilon_1 & \varepsilon_2 \\ \varepsilon_2 & \varepsilon_1 & 1-\varepsilon_1-\varepsilon_2 \end{bmatrix}$$

- Special case

$$p(y|x) = \begin{bmatrix} 1-\varepsilon & \varepsilon & 0 \\ 0 & \varepsilon & 1-\varepsilon \end{bmatrix}$$



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Channel Model

□ Symmetric Channel

- Three definitions:
- The rows of the probability transition matrix are **置换** permutations of each other and so are the columns. → symmetric I
- The rows of the probability transition matrix are permutations of each other and all the column sums are equal → weakly symmetric
- (The columns can be partitioned into subsets in such a way that in each subset, the rows are permutations of each other and so are the columns. → symmetric II)

$$\begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \end{bmatrix} \quad \begin{pmatrix} \frac{1}{3} & \frac{1}{6} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{pmatrix} \quad \begin{matrix} 0 & 1 & 2 \\ 0 & 1 \\ 1 & \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.7 \end{bmatrix} \end{matrix}$$

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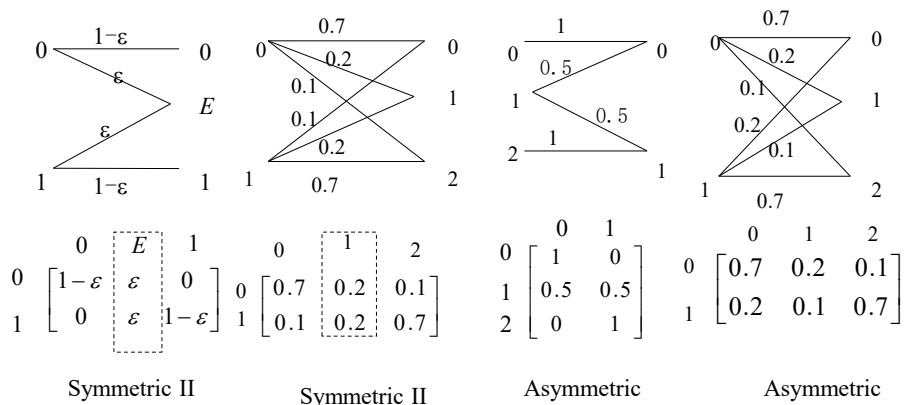
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Channel Model

□ Some examples



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Channel Capacity

- **Definition** We define the “information” channel capacity of a discrete memoryless channel as

$$C = \max_{p(x)} I(X; Y), \quad (7.1)$$

where the maximum is taken over all possible input distributions $p(x)$.

□ **Interpretation**

- Mutual information: A measure of the amount of information that one random variable contains about another random variable.
 - Original uncertainty (information): $H(X)$
 - Uncertainty at the receiver (Y observed): $H(X|Y)$
 - Uncertainty reduction: $H(X) - H(X|Y) \triangleq I(X; Y)$
 - What is the largest average amount of information can be sent through the channel every time.
- Input alphabet $p(x)$
- How noisy the channel is

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Channel Capacity

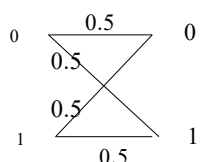
□ **Channel capacity examples**

- Noiseless binary channel

$$\begin{array}{ccc} 0 & \longrightarrow & 0 \\ 1 & \longrightarrow & 1 \end{array} \quad C = \max I(X; Y) = \max I(X; X) = \max H(X) = 1$$

$$p(x=0) = p(x=1) = \frac{1}{2}$$

- Strong noisy channel



$$p(y|x) = p(y)$$

$$\begin{aligned} I(X; Y) &= \sum_{x,y} p(x,y) \log \frac{p(y|x)}{p(y)} \\ &= \sum_{x,y} p(x,y) \log \frac{p(y)}{p(y)} \\ &= \sum_{x,y} p(x,y) \log 1 \\ &= 0 \end{aligned}$$

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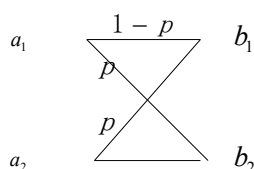
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Channel Capacity



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General Binary Symmetric Channel (BSC)



$$\begin{aligned}
 I(X; Y) &= H(Y) - H(Y|X) \\
 &= H(Y) - \sum p(x) H(Y|X = x) \\
 &= H(Y) - \sum p(x) H(p) \\
 &= H(Y) - H(p) \\
 &\leq 1 - H(p),
 \end{aligned}$$

Equality is achieved when the input distribution is uniform.

$$X = \begin{cases} 1 & \text{with probability } p, \\ 0 & \text{with probability } 1 - p. \end{cases}$$

$$H(p) \stackrel{\text{def}}{=} -p \log p - (1 - p) \log(1 - p)$$

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Channel Capacity



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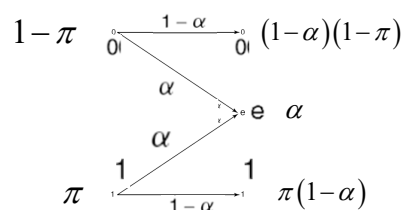
Binary Erasure Channel

$$\begin{aligned}
 C &= \max_{p(x)} I(X; Y) \\
 &= \max_{p(x)} (H(Y) - H(Y|X)) \\
 &= \max_{p(x)} H(Y) - H(\alpha).
 \end{aligned}$$

How to solve $\max_{p(x)} H(Y)$?

letting $\Pr(X = 1) = \pi$

$$\begin{aligned}
 H(Y) &= H((1 - \pi)(1 - \alpha), \alpha, \pi(1 - \alpha)) \\
 &= H(\alpha) + (1 - \alpha)H(\pi).
 \end{aligned}$$



$$\begin{aligned}
 C &= \max_{p(x)} H(Y) - H(\alpha) \\
 &= \max_{\pi} (1 - \alpha)H(\pi) + H(\alpha) - H(\alpha) \\
 &= \max_{\pi} (1 - \alpha)H(\pi) \\
 &= 1 - \alpha,
 \end{aligned}$$

where capacity is achieved by $\pi = \frac{1}{2}$.

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- The expression for the capacity has some intuitive meaning:
- Since a proportion α of the bits are lost in the channel, we can recover (at most) a proportion $1 - \alpha$ of the bits. Hence the capacity is at most $1 - \alpha$.

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Channel Capacity

- Channel capacity for a general discrete memoryless channel (一般离散无记忆信道的容量计算)

$$\begin{aligned}
 C &\triangleq \max_{p(x)} I(X;Y) && X \longrightarrow \boxed{P(Y|X)} \longrightarrow Y \\
 &= \max_{p(x)} \sum_{x,y} p(x,y) \log \frac{p(y|x)}{p(y)} \\
 &= \max_{p(x)} \sum_{x,y} p(x,y) \log \frac{p(y|x)}{\sum_x p(x)p(y|x)} \\
 &= \max_{p(x)} \sum_{x,y} p(x)p(y|x) \log \frac{p(y|x)}{\sum_x p(x)p(y|x)}
 \end{aligned}$$

We have to do this optimization over $p(x)$ to find the channel capacity

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Channel Capacity



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- The optimization is actually to find the maximum of a concave function (凹函数的最大)
- Recall

Theorem 2.7.4 Let $(X, Y) \sim p(x, y) = p(x)p(y|x)$. The mutual information $I(X; Y)$ is a concave function of $p(x)$ for fixed $p(y|x)$

- If we can solve this optimization problem, then we find the closed-form solution (闭式解) for the capacity.
- In general, there is no closed-form solution for the capacity. But for many simple channels it is possible to calculate the capacity using properties such as **symmetry**. Some of the examples considered earlier are of this form.
- 一般信道的容量计算没有闭式解，对称信道非常特殊

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Channel Capacity



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- Channel Capacity of Symmetric Channel (对称信道的容量)

Theorem 7.2.1 For a weakly symmetric channel,

$$C = \log |\mathcal{Y}| - H(\text{row of transition matrix}),$$

and this is achieved by a uniform distribution on the input alphabet.

Letting $\mathbf{r}$ be a row of the transition matrix, we have

$$I(X; Y) = H(Y) - H(Y|X) = H(Y) - H(\mathbf{r}) \leq \log |\mathcal{Y}| - H(\mathbf{r})$$

with equality if the output distribution is uniform.

But $p(x) = 1/|\mathcal{X}|$ achieves a uniform distribution on $\mathcal{Y}$, as seen from

$$p(y) = \sum_{x \in \mathcal{X}} p(y|x)p(x) = \frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} p(y|x) = c \frac{1}{|\mathcal{X}|} = \frac{1}{|\mathcal{Y}|}$$

where c is the sum of the entries in one column of the probability transition matrix.

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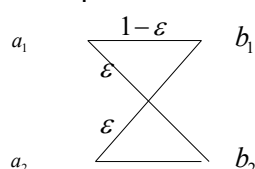
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Channel Capacity

□ Example



$$\begin{aligned}
 I(X; Y) &= H(Y) - H(Y|X) \\
 &= H(Y) - \sum p(x) H(Y|X = x) \\
 &= H(Y) - \sum p(x) H(p) \\
 &= H(Y) - H(p) \\
 &\leq 1 - H(p),
 \end{aligned}$$

$$\begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0.3 & 0.2 \\ 0.2 & 0.5 & 0.3 \end{bmatrix}$$

$$\begin{pmatrix} \frac{1}{3} & \frac{1}{6} & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{pmatrix}$$

$$C = \max_{p(x)} I(X; Y) = \log 3 - H(0.5, 0.3, 0.2),$$

□ The above theorem also holds for *symmetric II channel*.

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Channel Capacity

□ Properties of channel capacity

1. $C \geq 0$ since $I(X; Y) \geq 0$.
2. $C \leq \log |\mathcal{X}|$ since $C = \max I(X; Y) \leq \max H(X) = \log |\mathcal{X}|$.
3. $C \leq \log |\mathcal{Y}|$ for the same reason.
4. $I(X; Y)$ is a continuous function of $p(x)$.

□ The properties of channel capacity actually comes from those of Mutual information. (信道容量就是互信息)

□ Since $I(X; Y)$ is a concave function over a closed convex set, a local maximum is a **global maximum**. From properties 2 and 3, the maximum is finite, and we are justified in using the term *maximum* rather than *supremum* (上确界) in the definition of capacity. The maximum can then be found by standard **nonlinear optimization techniques such as gradient search**.

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Channel Capacity

$$C \triangleq \max_{p(x)} I(X;Y) = \max_{p(x)} \sum_{x,y} p(x)p(y|x) \log \frac{p(y|x)}{\sum_x p(x)p(y|x)}$$

$$\frac{\partial I(X;Y)}{\partial p(x_n)} = \frac{\partial}{\partial p(x_n)} \sum_{x_n, y_j} p(y_j|x_n)p(x_n) \log \frac{p(y_j|x_n)}{p(y_j)}$$

=

$$= \sum_j p(y_j|x_n) \log \frac{p(y_j|x_n)}{p(y_j)} - \log e$$

$$= I(X=x_n;Y) - \log e$$

Partial Mutual Information

$$I(X=x_n;Y) = \sum_j p(y_j|x_n) \log \frac{p(y_j|x_n)}{p(y_j)}$$

A set of **necessary and sufficient conditions** on an input probability vector $p(x_1), p(x_2), \dots, p(x_n)$ to achieve capacity on a DMC is

$$I(X=x_n;Y) = C, \quad p(x_n) > 0$$

$$I(X=x_n;Y) \leq C, \quad p(x_n) = 0$$

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- The *information channel capacity* does not give any intuition of how to transmit information to achieve the channel capacity.
- In the following we will investigate this issue using *operational channel capacity*.



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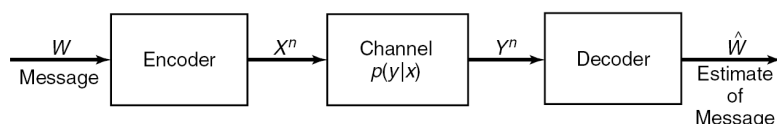
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Channel Coder and Decoder



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□ A communication system



A message W , drawn from the index set $\{1, 2, \dots, M\}$, results in the signal $X^n(W)$, which is received by the receiver as a random sequence $Y^n \sim p(y^n|x^n)$. The receiver then guesses the index W by an appropriate decoding rule $\hat{W} = g(Y^n)$. The receiver makes an error if $\hat{W}$ is not the same as the index W that was transmitted.

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Channel Coder and Decoder



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Definition A discrete channel, denoted by $(\mathcal{X}, p(y|x), \mathcal{Y})$, consists of two finite sets $\mathcal{X}$ and $\mathcal{Y}$ and a collection of probability mass functions $p(y|x)$, one for each $x \in \mathcal{X}$, such that for every x and y , $p(y|x) \geq 0$, and for every x , $\sum_y p(y|x) = 1$, with the interpretation that X is the input and Y is the output of the channel.

Definition The n th extension of the discrete memoryless channel (DMC) is the channel $(\mathcal{X}^n, p(y^n|x^n), \mathcal{Y}^n)$, where

$$p(y_k|x^k, y^{k-1}) = p(y_k|x_k), \quad k = 1, 2, \dots, n.$$

So that

$$p(y^n|x^n) = \prod_{i=1}^n p(y_i|x_i)$$

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Channel Coder and Decoder

□ Channel Coder and Decoder

Definition An (M, n) code for the channel $(\mathcal{X}, p(y|x), \mathcal{Y})$ consists of the following:

1. An index set $\{1, 2, \dots, M\}$.
2. An encoding function $X^n: \{1, 2, \dots, M\} \rightarrow \mathcal{X}^n$, yielding codewords $x^n(1), x^n(2), \dots, x^n(M)$. The set of codewords is called the *codebook*.
3. A decoding function

$$g: \mathcal{Y}^n \rightarrow \{1, 2, \dots, M\},$$

which is a deterministic rule that assigns a guess to each possible received vector.

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Channel Coder and Decoder

□ Probability of error

Definition (*Conditional probability of error*) Let

$$\lambda_i = \Pr(g(Y^n) \neq i | X^n = x^n(i)) = \sum_{y^n} p(y^n | x^n(i)) I(g(y^n) \neq i)$$

be the *conditional probability of error* given that index i was sent, where $I(\cdot)$ is the indicator function.

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Channel Coder and Decoder



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Definition The (arithmetic) average probability of error $P_e^{(n)}$ for an (M, n) code is defined as

算数平均
$$P_e^{(n)} = \frac{1}{M} \sum_{i=1}^M \lambda_i.$$

If the index W is chosen uniformly over the set $\{1, 2, \dots, M\}$,

$$P_e^{(n)} \triangleq \Pr(W \neq g(Y^n)),$$

$$\sum_s P(e|s)P(s)$$

Definition The rate R of an (M, n) code is

$$R = \frac{\log M}{n} \quad \text{bits per transmission.}$$

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Channel Coder and Decoder



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Definition A rate R is said to be *achievable* if there exists a sequence of $(\lceil 2^{nR} \rceil, n)$ codes such that the maximal probability of error $\lambda^{(n)}$ tends to 0 as $n \rightarrow \infty$.

Later, we write $(2^{nR}, n)$ codes to mean $(\lceil 2^{nR} \rceil, n)$ codes. This will simplify the notation.

Definition The *capacity* of a channel is the supremum of all achievable rates.

Thus, rates less than capacity yield arbitrarily small probability of error for sufficiently large block lengths.

Operational channel capacity: Shannon's second theorem establishes that the information channel capacity is equal to the operational channel capacity.

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Joint Typical Set

□ Joint Typical Set

Definition The set $A_\epsilon^{(n)}$ of *jointly typical* sequences $\{(x^n, y^n)\}$ with respect to the distribution $p(x, y)$ is the set of n -sequences with empirical entropies ϵ -close to the true entropies:

$$A_\epsilon^{(n)} = \{(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n :$$

$$X \text{ typical} \longrightarrow \left| -\frac{1}{n} \log p(x^n) - H(X) \right| < \epsilon,$$

$$Y \text{ typical} \longrightarrow \left| -\frac{1}{n} \log p(y^n) - H(Y) \right| < \epsilon,$$

$$\text{Joint relationship} \longrightarrow \left| -\frac{1}{n} \log p(x^n, y^n) - H(X, Y) \right| < \epsilon \},$$

where

$$p(x^n, y^n) = \prod_{i=1}^n p(x_i, y_i).$$

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Joint Typical Set

□ Properties

Theorem 7.6.1 (Joint AEP) Let (X^n, Y^n) be sequences of length n drawn i.i.d. according to $p(x^n, y^n) = \prod_{i=1}^n p(x_i, y_i)$. Then:

$$1. \Pr((X^n, Y^n) \in A_\epsilon^{(n)}) \rightarrow 1 \text{ as } n \rightarrow \infty.$$

Proof

$$\Pr\left(\left| -\frac{1}{n} \log p(X^n) - H(X) \right| \geq \epsilon\right) < \frac{\epsilon}{3}.$$

Weak law of
large numbers

$$\Pr\left(\left| -\frac{1}{n} \log p(Y^n) - H(Y) \right| \geq \epsilon\right) < \frac{\epsilon}{3}$$

$$P\left(\left| \frac{X_1 + X_2 + \dots + X_n}{n} - \mu \right| \geq \epsilon\right) \xrightarrow{n \rightarrow \infty} 0 \quad \Pr\left(\left| -\frac{1}{n} \log p(X^n, Y^n) - H(X, Y) \right| \geq \epsilon\right) < \frac{\epsilon}{3}.$$

The probability of the union of the sets must be less than ϵ

Hence for n sufficiently large, the probability of the joint typical set is greater than $1 - \epsilon$

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Joint Typical Set

□ Properties

Theorem 7.6.1 (Joint AEP) Let (X^n, Y^n) be sequences of length n drawn i.i.d. according to $p(x^n, y^n) = \prod_{i=1}^n p(x_i, y_i)$. Then:

1. $\Pr((X^n, Y^n) \in A_\epsilon^{(n)}) \rightarrow 1$ as $n \rightarrow \infty$.
2. $|A_\epsilon^{(n)}| \leq 2^{n(H(X,Y)+\epsilon)}$.

Proof

$$\begin{aligned}
 1 &= \sum p(x^n, y^n) \\
 &\geq \sum_{(x^n, y^n) \in A_\epsilon^{(n)}} p(x^n, y^n) & |A_\epsilon^{(n)}| &\leq 2^{n(H(X,Y)+\epsilon)} \\
 &\geq |A_\epsilon^{(n)}| 2^{-n(H(X,Y)+\epsilon)},
 \end{aligned}$$

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Joint Typical Set

□ Properties

3. If $(\tilde{X}^n, \tilde{Y}^n) \sim p(x^n)p(y^n)$ [i.e., $\tilde{X}^n$ and $\tilde{Y}^n$ are independent with the same marginals as $p(x^n, y^n)$], then

$$\Pr((\tilde{X}^n, \tilde{Y}^n) \in A_\epsilon^{(n)}) \leq 2^{-n(I(X;Y)-3\epsilon)}.$$

Also, for sufficiently large n ,

$$\Pr((\tilde{X}^n, \tilde{Y}^n) \in A_\epsilon^{(n)}) \geq (1 - \epsilon) 2^{-n(I(X;Y)+3\epsilon)}.$$

Proof

$$\begin{aligned}
 \Pr((\tilde{X}^n, \tilde{Y}^n) \in A_\epsilon^{(n)}) &= \sum_{(x^n, y^n) \in A_\epsilon^{(n)}} \underbrace{p(x^n)}_{\leq 2^{n(H(X,Y)+\epsilon)}} \underbrace{p(y^n)}_{\leq 2^{-n(H(X)-\epsilon)} 2^{-n(H(Y)-\epsilon)}} \\
 &\leq 2^{n(H(X,Y)+\epsilon)} 2^{-n(H(X)-\epsilon)} 2^{-n(H(Y)-\epsilon)} \\
 &= 2^{-n(I(X;Y)-3\epsilon)}.
 \end{aligned}$$

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Mutual Information

Relationship (Venn diagram)

Theorem 2.4.1 (Mutual information and entropy)

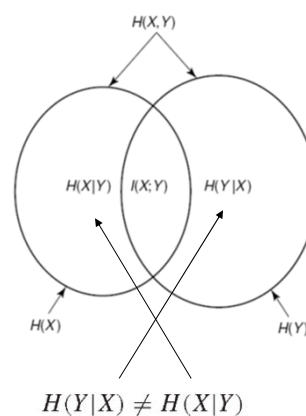
$$I(X; Y) = H(X) - H(X|Y)$$

$$I(X; Y) = H(Y) - H(Y|X)$$

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

$$I(X; Y) = I(Y; X)$$

$$I(X; X) = H(X).$$



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Joint Typical Set

For sufficiently large n , $\Pr(A_\epsilon^{(n)}) \geq 1 - \epsilon$, and therefore

$$\begin{aligned} 1 - \epsilon &\leq \sum_{(x^n, y^n) \in A_\epsilon^{(n)}} p(x^n, y^n) \\ &\leq |A_\epsilon^{(n)}| 2^{-n(H(X, Y) - \epsilon)} \quad |A_\epsilon^{(n)}| \geq (1 - \epsilon) 2^{n(H(X, Y) - \epsilon)} \end{aligned}$$

$$\begin{aligned} \Pr((\tilde{X}^n, \tilde{Y}^n) \in A_\epsilon^{(n)}) &= \sum_{A_\epsilon^{(n)}} p(x^n) p(y^n) \\ &\geq (1 - \epsilon) 2^{n(H(X, Y) - \epsilon)} 2^{-n(H(X) + \epsilon)} 2^{-n(H(Y) + \epsilon)} \\ &= (1 - \epsilon) 2^{-n(I(X; Y) + 3\epsilon)}. \quad \square \end{aligned}$$

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Channel Coding Theorem

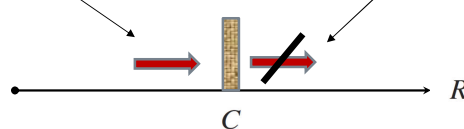
□ Channel Coding Theorem

Theorem 7.7.1 (Channel coding theorem) For a discrete memoryless channel, all rates below capacity C are achievable. Specifically, for every rate $R < C$, there exists a sequence of $(2^n, n)$ codes with maximum probability of error $\lambda^{(n)} \rightarrow 0$.

Conversely, any sequence of $(2^n, n)$ codes with $\lambda^{(n)} \rightarrow 0$ must have $R \leq C$.

Proof: All operational capacity proof can be divided into two parts:

- i) Achievability: All $R \leq C$ are achievable
- ii) Converse: No rate $R > C$ is achievable



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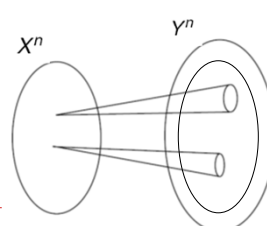
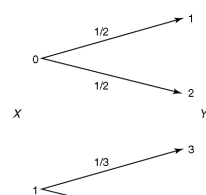
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Joint Typical Set

□ Decoding strategy

- The basic idea is that for large block lengths, every channel has a subset of inputs that produce essentially disjoint sequences at the output.
- Joint typical decoding
- we decode a channel output Y^n as the i th index if the codeword $X^n(i)$ is "jointly typical" with the received signal Y^n .

For each (typical) input n -sequence, there are approximately $2^{nH(\hat{Y}|X)}$ possible Y



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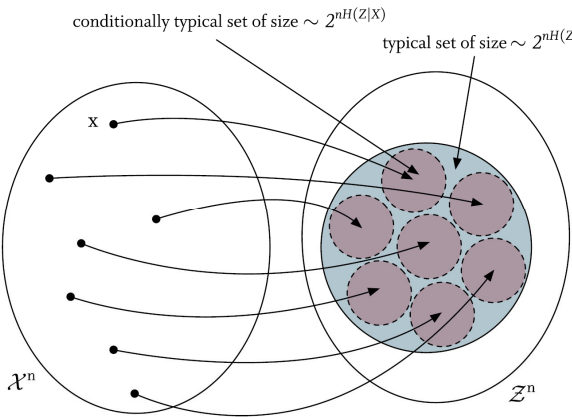
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conditionally typical set of size $\sim 2^{nH(Z|X)}$

typical set of size $\sim 2^{nH(Z)}$

X^n Z^n

The total number of possible (typical) Y sequences is $\approx 2^{nH(Y)}$. This set has to be divided into sets of size $2^{nH(Y|X)}$ corresponding to the different input X sequences. The total number of disjoint sets is less than or equal to $2^{n(H(Y)-H(Y|X))} = 2^{nI(X;Y)}$. Hence, we can send at most $\approx 2^{nI(X;Y)}$ distinguishable sequences of length n .

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Channel Coding Theorem

- Shannon's second theorem -- *Channel coding theorem*
 - The achievability issue of the channel capacity
- Intuition (步骤)
 - 1, Random coding, joint typicality decoding
 - 2, calculate the average of the probability of error
 - 3, Allowing an arbitrarily small but nonzero probability of error

With high probability the transmitted codeword and the received sequence are **jointly typical**, since they are probabilistically related.

We **look for a codeword that is jointly typical with the received sequence**. If we find a unique one, we declare that codeword to be the transmitted one.

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Channel Coding Theorem

□ Achievability

- Generate a $(2^{nR}, n)$ code at **random** according to the distribution $p(x)$

$$\mathcal{C} = \begin{bmatrix} x_1(1) & x_2(1) & \cdots & x_n(1) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(2^{nR}) & x_2(2^{nR}) & \cdots & x_n(2^{nR}) \end{bmatrix}, \quad p(x^n) = \prod_{i=1}^n p(x_i)$$

$$\Pr(\mathcal{C}) = \prod_{w=1}^{2^{nR}} \prod_{i=1}^n p(x_i(w)).$$

Each entry in this matrix is **generated i.i.d.**

- The code $\mathcal{C}$ is then revealed to both sender and receiver. Both sender and receiver are also assumed to know the channel transition matrix $p(y|x)$ for the channel. (收发双端都已知码书和信道)
- A message W is chosen according to a uniform distribution

$$\Pr(W = w) = 2^{-nR}, \quad w = 1, 2, \dots, 2^{nR}.$$

- The codeword corresponding to the w -th row of $\mathcal{C}$, is sent over the channel. (每一行是一个codeword码字)

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Channel Coding Theorem

- The receiver receives a sequence Y^n according to the distribution

$$P(y^n|x^n(w)) = \prod_{i=1}^n p(y_i|x_i(w))$$

- The receiver guesses which message $\hat{W}$ was sent **using jointly typical decoding**
 - $(X^n(\hat{W}), Y^n)$ is jointly typical.
 - There is no other index $W' \neq \hat{W}$ such that $(X^n(W'), Y^n) \in A_\epsilon^{(n)}$
- JTD is not optimal to minimize probability of error, but is asymptotically optimal
- Easy to analyze
- **If no such $\hat{W}$ exists or if there is more than one such, an error is declared.**
- There is a decoding error if $\hat{W} \neq W$, Let $\mathcal{E}$ be the event $\{\hat{W} \neq W\}$

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Channel Coding Theorem

□ Analysis of the probability of error

- Instead of calculating the probability of error for a single code, we calculate the average over all codes generated at random according to the distribution. (分析计算平均错误)
- For a typical codeword, two kinds of error :
 - Either the output Y^n is not jointly typical with the transmitted codeword
 - there is some other codeword that is jointly typical with Y^n .
- For any rival codeword, the probability that it is jointly typical with the received sequence is approximately 2^{-nI} .

$$\Pr(\mathcal{E}) = \sum_c \underbrace{\Pr(C)}_{\text{averaged over all codebooks}} \underbrace{P_e^{(n)}(C)}_{\text{averaged over all codewords in the codebook}} = \sum_c \Pr(C) \frac{1}{2^{nR}} \sum_{w=1}^{2^{nR}} \lambda_w(C) = \frac{1}{2^{nR}} \sum_{w=1}^{2^{nR}} \sum_c \Pr(C) \lambda_w(C),$$

$\Pr(g(Y^n) \neq i | X^n = x^n(i))$
发w的条件错误概率

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Channel Coding Theorem

By the symmetry of the code construction, the average probability of error averaged over all codes does not depend on the particular index that was sent.

Thus, we can assume without loss of generality that the message $W = 1$ was sent

$$= \frac{1}{2^{nR}} \sum_{w=1}^{2^{nR}} \sum_c \Pr(C) \lambda_w(C) = \sum_c \Pr(C) \lambda_1(C) = \Pr(\mathcal{E} | W = 1).$$

Define the events: $E_i = \{ (X^n(i), Y^n) \text{ is in } A_\epsilon^{(n)} \}, \quad i \in \{1, 2, \dots, 2^{nR}\},$

Error scenario:

- 1) When the transmitted codeword and the received sequence are not jointly typical
- 2) When a wrong codeword is jointly typical with the received sequence

$$\Pr(\mathcal{E} | W = 1) = P(E_1^c \cup E_2 \cup E_3 \cup \dots \cup E_{2^{nR}} | W = 1)$$

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Channel Coding Theorem

- By joint AEP (Property 1)

$$P(E_1^c | W = 1) \rightarrow 0, \rightarrow P(E_1^c | W = 1) \leq \epsilon$$

- By the code generation process, $X^n(1)$ and $X^n(i)$ are independent for $i \neq 1$. Hence, the probability that $X^n(i)$ and Y^n are jointly typical is $\leq 2^{-n(I(X;Y)-3\epsilon)}$ by the joint AEP (Property 3)

$$\begin{aligned} \Pr(\mathcal{E}) &= \Pr(\mathcal{E} | W = 1) \leq P(E_1^c | W = 1) + \sum_{i=2}^{2^{nR}} P(E_i | W = 1) \\ &\leq \epsilon + \sum_{i=2}^{2^{nR}} 2^{-n(I(X;Y)-3\epsilon)} \\ &= \epsilon + (2^{nR} - 1) 2^{-n(I(X;Y)-3\epsilon)} \\ &\leq \epsilon + 2^{3n\epsilon} 2^{-n(I(X;Y)-R)} \leq 2\epsilon \quad R < I(X;Y) - 3\epsilon. \end{aligned}$$

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Channel Coding Theorem

$$\Pr(\mathcal{E}) \xrightarrow{n \rightarrow \infty} 0 \quad \text{if } R < I(X;Y)$$

- Choose $p(x) \rightarrow p^*(x)$ that maximize $I(X;Y)$

$$R < \max_{p(x)} I(X;Y) = C$$

- Get rid of the average over codebooks. Since the average probability of error over codebooks is small, there exists at least one codebook $\mathcal{C}^*$ with a small average probability of error.

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Channel Coding Theorem



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□ Comments

- Although the theorem shows that there exist good codes with arbitrarily small probability of error for long block lengths, it does not provide a way of constructing the best codes. (没有给出如何构建最优码)
- If we used the scheme suggested by the proof and generate a code at random with the appropriate distribution, the code constructed **is likely to be** good for long block lengths. (码长很长时, 大概率不错的码)
- However, without some structure in the code, it is very difficult to decode (the simple scheme of table lookup requires an exponentially large table). Hence the theorem does not provide a practical coding scheme. (没有结构, 不实用)
- Ever since Shannon's original paper, researchers have tried to develop structured codes that are easy to encode and decode. a variety of techniques have been used to construct error correcting codes.
- Advanced courses; Error correct coding (channel coding)

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Channel Coding Theorem



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□ The Converse to the coding theorem (逆定理)

□ Fano's Inequality (费诺不等式)

- Suppose that we know a random variable Y and we wish to guess the value of a correlated random variable X . Fano's inequality relates the probability of error in guessing the random variable X to its conditional entropy $H(X|Y)$
- 重要意义: 把根据 Y 猜测 X 的错误概率跟条件熵关联起来。
- $X \sim p(x)$, Y is related to X by the conditional distribution $p(y|x)$. From Y , we calculate an estimate $g(Y) = \hat{X}$ taking on values in $\mathcal{X}$. We wish to bound the probability that $\hat{X} \neq X$.
- We observe that $X \rightarrow Y \rightarrow \hat{X}$ forms a Markov chain.
- Define the probability of error

$$P_e = \Pr \{ \hat{X} \neq X \}$$

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Channel Coding Theorem

□ Fano's Inequality

Theorem 2.10.1 (Fano's Inequality) For any estimator $\hat{X}$ such that $X \rightarrow Y \rightarrow \hat{X}$, with $P_e = \Pr(X \neq \hat{X})$, we have

$$H(P_e) + P_e \log |\mathcal{X}| \geq H(X|\hat{X}) \geq H(X|Y)$$

This inequality can be weakened to $1 + P_e \log |\mathcal{X}| \geq H(X|Y)$

Proof: Define an error random variable,

$$E = \begin{cases} 1 & \text{if } \hat{X} \neq X, \\ 0 & \text{if } \hat{X} = X. \end{cases} \quad \text{a binary-valued random variable}$$

$$H(X|\hat{X}) + H(E|X, \hat{X}) = \underbrace{H(E, X|\hat{X})}_{=0} = H(E|\hat{X}) + H(X|E, \hat{X}).$$

chain rule for Joint entropies

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Channel Coding Theorem

$$\Rightarrow H(E|\hat{X}) \leq H(E) = H(P_e) \quad E \text{ is a binary-valued random variable.}$$

Conditioning reduce entropy

$$\Rightarrow H(X|E, \hat{X}) = \Pr(E=0)H(X|\hat{X}, E=0) + \Pr(E=1)H(X|\hat{X}, E=1) \\ \leq (1 - P_e)0 + P_e \log |\mathcal{X}|,$$

Upper bound the conditional entropy by the log of the number of possible outcomes.

$$H(X|\hat{X}) = H(E|\hat{X}) + H(X|E, \hat{X}) \\ \leq H(P_e) + P_e \log |\mathcal{X}|$$

By the data-processing inequality, we have $I(X; \hat{X}) \leq I(X; Y)$ since $X \rightarrow Y \rightarrow \hat{X}$ is a Markov chain, and therefore $H(X|\hat{X}) \geq H(X|Y)$. Thus,

$$H(P_e) + P_e \log |\mathcal{X}| \geq H(X|\hat{X}) \geq H(X|Y)$$

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Channel Coding Theorem

Lemma 7.9.1 (Fano's inequality) For a discrete memoryless channel with a codebook $\mathcal{C}$ and the input message W uniformly distributed over 2^{nR} , we have

$$H(W|\hat{W}) \leq 1 + P_e^{(n)} nR.$$

Proof:

Since W is uniformly distributed, we have $P_e^{(n)} = \Pr(W \neq \hat{W})$. We apply Fano's inequality for W in an alphabet of size 2^{nR} .

We will now prove a lemma which shows that the capacity per transmission is not increased if we use a discrete memoryless channel many times.

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Channel Coding Theorem

□ The capacity per transmission is not increased if we use a discrete memoryless channel many times.

Lemma 7.9.2 Let Y^n be the result of passing X^n through a discrete memoryless channel of capacity C . Then

$$I(X^n; Y^n) \leq nC \quad \text{for all } p(x^n).$$

Proof $I(X^n; Y^n) = H(Y^n) - H(Y^n|X^n)$

$$= H(Y^n) - \sum_{i=1}^n H(Y_i|Y_1, \dots, Y_{i-1}, X^n)$$

$$= H(Y^n) - \sum_{i=1}^n H(Y_i|X_i),$$

$$\leq \sum_{i=1}^n H(Y_i) - \sum_{i=1}^n H(Y_i|X_i) = \sum_{i=1}^n I(X_i; Y_i) \leq nC,$$

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Channel Coding Theorem

□ The Converse to the coding theorem (逆定理)

- We have to show that any sequence of $(2^{nR}, n)$ codes with $\lambda^{(n)} \rightarrow 0$ must have $R \leq C$.

we have $W \rightarrow X^n(W) \rightarrow Y^n \rightarrow \hat{W}$.

let W be drawn according to a uniform distribution $\Pr(\hat{W} \neq W) = P_e^{(n)} = \frac{1}{2^{nR}} \sum_i \lambda_i$

$$nR \stackrel{(a)}{=} H(W)$$

$$\stackrel{(b)}{=} H(W|\hat{W}) + I(W; \hat{W})$$

follows from the assumption
that W is uniform over

$$\stackrel{(c)}{\leq} 1 + P_e^{(n)} nR + I(W; \hat{W}) \rightarrow \text{is Fano's inequality for } W \text{ taking on at most } 2^{nR} \text{ values}$$

$$\stackrel{(d)}{\leq} 1 + P_e^{(n)} nR + I(X^n; Y^n) \rightarrow \text{Is the data-processing inequality}$$

$$\stackrel{(e)}{\leq} 1 + P_e^{(n)} nR + nC, \quad \rightarrow \text{is from Lemma 7.9.2.}$$

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Channel Coding Theorem

$$R \leq P_e^{(n)} R + \frac{1}{n} + C.$$

$$n \rightarrow \infty \quad \begin{matrix} \searrow & \searrow \\ 0 & 0 \end{matrix} \quad R \leq C.$$

$$P_e^{(n)} \geq 1 - \frac{C}{R} - \frac{1}{nR}.$$

This equation shows that if $R > C$, the probability of error is bounded away from 0 for sufficiently large n

Hence, we cannot achieve an arbitrarily low probability of error at rates above capacity.

This converse is sometimes called the **weak converse** to the channel coding theorem. It is also possible to prove a **strong converse**, which states that for rates above capacity, the probability of error goes exponentially to 1.

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SOURCE-CHANNEL SEPARATION THEOREM (信源-信道分离定理)



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- Data compression ($L > H$: Theorem 5.4.2)
- Data transmission ($R < C$: Theorem 7.7.1).

Theorem 5.4.2 The minimum expected codeword length per symbol satisfies

$$\frac{H(X_1, X_2, \dots, X_n)}{n} \leq L_n^* < \frac{H(X_1, X_2, \dots, X_n)}{n} + \frac{1}{n}.$$

Moreover, if $X_1, X_2, \dots, X_n$ is a stationary stochastic process,

$$L_n^* \rightarrow H(\mathcal{X}),$$

where $H(\mathcal{X})$ is the entropy rate of the process.

Theorem 7.7.1 (Channel coding theorem) For a discrete memoryless channel, all rates below capacity C are achievable. Specifically, for every rate $R < C$, there exists a sequence of $(2^{nR}, n)$ codes with maximum probability of error $\lambda^{(n)} \rightarrow 0$.

Conversely, any sequence of $(2^{nR}, n)$ codes with $\lambda^{(n)} \rightarrow 0$ must have
2025/3: $R \leq C$.

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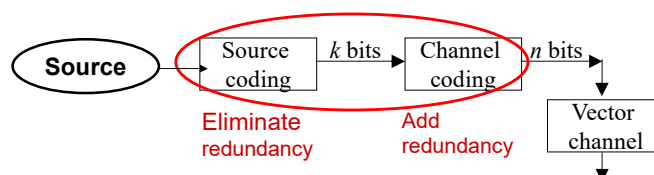
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SOURCE-CHANNEL SEPARATION THEOREM (信源-信道分离定理)



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- For example, consider sending digitized speech or music over a discrete memoryless channel. We could
- Design a code to map the sequence of speech samples directly into the input of the channel,
- or We could compress the speech into its most efficient representation, then use the appropriate channel code to send it over the channel.



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SOURCE-CHANNEL SEPARATION THEOREM (信源-信道分离定理)



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- It is not immediately clear that we are not losing something by using the two-stage method, since data compression does not depend on the channel and the channel coding does not depend on the source distribution.
- IF the two-stage method is as good as any other method of transmitting information over a noisy channel.
- It implies that we can consider the design of a communication system as a combination of two parts, source coding and channel coding. We can design source codes for the most efficient representation of the data. We can, separately and independently, design channel codes appropriate for the channel. The combination will be as efficient as anything we could design by considering both problems together.

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Joint source-channel coding theorem



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Theorem 7.13.1 (Source-channel coding theorem) *If $V_1, V_2, \dots, V^n$ is a finite alphabet stochastic process that satisfies the AEP and $H(\mathcal{V}) < C$, there exists a source-channel code with probability of error $\Pr(\hat{V}^n \neq V^n) \rightarrow 0$. Conversely, for any stationary stochastic process, if $H(\mathcal{V}) > C$, the probability of error is bounded away from zero, and it is not possible to send the process over the channel with arbitrarily low probability of error.*

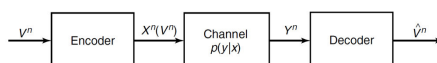


FIGURE 7.14. Joint source and channel coding.

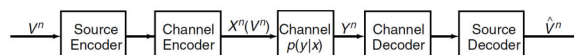


FIGURE 7.15. Separate source and channel coding.

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Joint source–channel coding theorem



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- Hence, we can transmit a stationary ergodic source over a channel if and only if its entropy rate is less than the capacity of the channel.
- The joint source–channel separation theorem enables us to consider the problem of source coding separately from the problem of channel coding. (分而治之)
- The source coder tries to find the most efficient representation of the source, and the channel coder encodes the message to combat the noise and errors introduced by the channel. The separation theorem says that the separate encoders (Figure 7.15) can achieve the same rates as the joint encoder (Figure 7.14).

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Remarks



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- The result—that a two-stage process is as good as any one-stage process—seems so obvious that it may be appropriate to point out that it is not always true.
- 并不是总是可分离的
- There are examples of multiuser channels where the decomposition breaks down.
- 多用户信道下，这种分离是不成立的

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Summary

□ Channel Model

- Channel transition matrix
- Discrete memoryless channel
 - BSC, BEC
 - Symmetric Channel

□ Channel Capacity

- Information channel capacity $C = \max_{p(x)} I(X; Y)$,
- The calculation a general DMC
 $I(X; Y)$ is a continuous concave function of $p(x)$.
- Channel capacity for a symmetric channel
 - Symmetric I II, weakly symmetric
 - Uniform input distribution achieves the capacity

□ Partial Mutual Information $I(X = x_n; Y) = \sum_j p(y_j | x_n) \log \frac{p(y_j | x_n)}{p(y_j)}$

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Summary

□ Joint typicality

$$A_\epsilon^{(n)} = \{(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n : \left| -\frac{1}{n} \log p(x^n) - H(X) \right| < \epsilon,$$

$$\left| -\frac{1}{n} \log p(y^n) - H(Y) \right| < \epsilon, \left| -\frac{1}{n} \log p(x^n, y^n) - H(X, Y) \right| < \epsilon \}$$

$$p(x^n, y^n) = \prod_{i=1}^n p(x_i, y_i).$$

□ Joint AEP

1. $\Pr((X^n, Y^n) \in A_\epsilon^{(n)}) \rightarrow 1$ as $n \rightarrow \infty$.
2. $|A_\epsilon^{(n)}| \leq 2^{n(H(X, Y) + \epsilon)}$.
3. If $(\tilde{X}^n, \tilde{Y}^n) \sim p(x^n)p(y^n)$ [i.e., $\tilde{X}^n$ and $\tilde{Y}^n$ are independent with the same marginals as $p(x^n, y^n)$], then

$$\Pr((\tilde{X}^n, \tilde{Y}^n) \in A_\epsilon^{(n)}) \leq 2^{-n(I(X, Y) - 3\epsilon)}.$$

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Summary

□ Channel Coding Theorem

- Operational channel capacity
- Joint typical decoding

Theorem 7.7.1 (Channel coding theorem) For a discrete memoryless channel, all rates below capacity C are achievable. Specifically, for every rate $R < C$, there exists a sequence of $(2^{nR}, n)$ codes with maximum probability of error $\lambda^{(n)} \rightarrow 0$.

Conversely, any sequence of $(2^{nR}, n)$ codes with $\lambda^{(n)} \rightarrow 0$ must have $R \leq C$.

- Achievability
- Converse
- Fano's inequality

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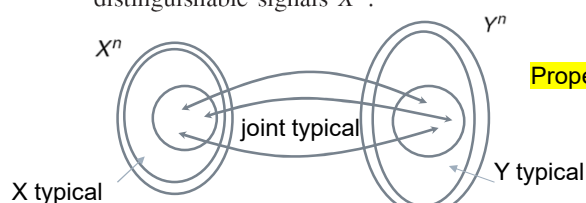
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□ Jointly (从联合的角度)

$2^{nH(X)}$ typical X sequences and about $2^{nH(Y)}$ typical Y sequences. However, since there are only $2^{nH(X,Y)}$ jointly typical sequences, not all pairs of typical X^n and typical Y^n are also jointly typical. The probability that any randomly chosen pair is jointly typical is about $2^{-nI(X;Y)}$. Hence, we can consider about $2^{nI(X;Y)}$ such pairs before we are likely to come across a jointly typical pair. This suggests that there are about $2^{nI(X;Y)}$ distinguishable signals X^n .



Property 3 of Joint Typical Set

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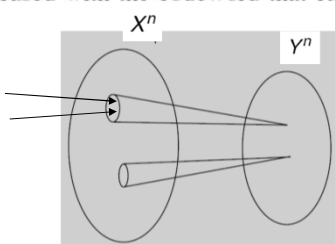
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Channel Coding Theorem

□ Look at the random encoder (从随机编码器的角度,符合证明过程)

For a received sequence Y^n , presumably the output sequence resulting from the true input signal X^n . For this sequence Y^n , there are about $2^{nH(X|Y)}$ conditionally typical input signals. The probability that some randomly chosen (other) input signal X^n is jointly typical with Y^n is about $2^{nH(X|Y)}/2^{nH(X)} = 2^{-nI(X;Y)}$. This again suggests that we can choose about $2^{nI(X;Y)}$ codewords $X^n(W)$ before one of these codewords will get confused with the codeword that caused the output Y^n .

不能出现两个
序列在同一个
条件典型集里



If we have fewer than 2^{nI} codewords, then with high probability there will be no other codewords that can be confused with the transmitted codeword, and the probability of error is small.

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